ADMM CASE STUDY

Data-Driven Retail Sales Optimization using Adventure Works

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I. INTRODUCTION

1.1 Introduction:

Data-Driven Retail Sales Optimization is a business intelligence and analytics solution developed using the Adventure Works sales dataset. The project focuses on analyzing sales performance, revenue trends, customer purchasing behavior, sales territory performance, product profitability, and sales channel contribution.

Adventure Works operates through two major sales channels: Internet sales, which represent direct customer transactions, and Reseller sales, which represent business-to-business transactions. Analyzing these channels together helps provide a consolidated view of overall sales performance.

The available datasets contain information about customers, employees, geographical locations, products, resellers, sales territories, dates, and sales transactions.

- Overall sales and revenue analysis
- Sales trends over time
- Internet and Reseller channel comparison
- Sales territory performance analysis
- Average order value analysis
- Customer segmentation by occupation and annual income
- Customer age-group analysis
- Identification of target customers under 50
- Product category sales analysis
- Gross profit and profitability margin analysis
- Interactive Power BI dashboards
- Data-driven business insights and recommendations

1.2 Business Problem & Objectives:

• Business Problem-

Retail businesses generate large amounts of sales and customer data. However, analyzing multiple datasets separately can make it difficult to obtain a consistent view of business performance.

Without an integrated analytical solution, it becomes challenging to identify:

- Overall sales and gross profit performance

- Revenue contribution from different sales channels
- Sales trends and year-over-year changes
- High-performing and underperforming territories
- Differences in average order value across territories
- Customer purchasing behavior by occupation and income
- Customer age groups with stronger sales performance
- Product categories that generate the highest revenue
- Product categories with the highest gross profit margins
- Opportunities for customer targeting and product optimization

The case study therefore focuses on developing an analytical pipeline that consolidates sales and dimension data into a centralized MySQL Data Warehouse and converts it into meaningful business insights through SQL analysis and Power BI dashboards.

• Objectives-

The key objectives of the project are:

- Load the available Adventure Works Excel datasets into Google Colab.
- Explore the source datasets using Python and Pandas.
- Profile record counts, data types, missing values, duplicates, and unique identifiers.
- Validate source records and dimensional keys.
- Clean and transform the source datasets.
- Prepare customer, employee, geography, product, reseller, sales territory, and date dimensions.
- Combine Internet and Reseller sales into a consolidated fact table.
- Export the transformed tables as CSV files.
- Create a structured MySQL Data Warehouse.
- Load the dimensional tables and consolidated sales fact table into MySQL.
- Validate primary keys, foreign keys, and record counts.
- Perform SQL-based sales and profitability analysis.
- Analyze sales trends and year-over-year growth.
- Compare Internet and Reseller sales performance.
- Identify high-performing and underperforming sales territories.
- Analyze customer sales by occupation and annual income.
- Analyze customer age groups and the under-50 customer segment.
- Evaluate product category sales, gross profit, and gross profit margin.
- Develop DAX measures for KPIs and analytical comparisons.
- Build interactive Power BI dashboards.
- Provide business insights and recommendations based on the observed sales patterns.

II. Architecture Overview

2.1 Architecture Overview

The Data-Driven Retail Sales Optimization solution follows a structured ETL, Data Warehouse, SQL Analytics, and Business Intelligence architecture.

The process begins with the source Adventure Works datasets and progresses through data exploration, validation, cleaning, transformation, dimensional modeling, database loading, SQL analysis, Power BI data modeling, DAX development, and dashboard creation.

The major components of the architecture are:

1. Source Data

The project uses 3 datasets:

- DimTables.xlsx
- FactInternetSales.xlsx
- FactResellerSales.xlsx

The dimension workbook contains customer, employee, geography, product, reseller, and sales territory information. The two fact workbooks contain Internet and Reseller sales transactions.

2. Python / Google Colab

Python and Pandas were used for:

- Loading Excel workbooks and sheets
- Data exploration and profiling
- Missing-value analysis
- Duplicate analysis
- Unique-key validation
- Data-type and date validation
- Data cleaning and transformation
- Foreign-key validation
- Combining Internet and Reseller sales
- Creating the date dimension
- Preparing analytical tables
- Exporting cleaned tables as CSV files

3. ETL & Cleaned Data

The source datasets were cleaned, validated, transformed, and prepared for dimensional modeling.

Major processing activities included:

- Standardizing column names and data types
- Validating customer, employee, product, reseller, and territory keys
- Removing three blank rows from the Reseller sales data
- Removing 41 Reseller sales records with invalid foreign keys
- Retaining appropriate channel-specific null values
- Removing the entirely blank CarrierTrackingNumber field from the consolidated fact table
- Preparing the date dimension
- Combining Internet and Reseller sales into fact_sales
- Exporting the cleaned dimension and fact tables

4. MySQL Data Warehouse

The transformed data was loaded into the MySQL database:

adventure_works_dw

The warehouse contains:

- dim_customer
- dim_employee
- dim_geography
- dim_product
- dim_reseller
- dim_sales_territory
- dim_date
- fact_sales

The dimension tables provide descriptive context for customers, employees, geography, products, resellers, sales territories, and dates. The consolidated fact table stores sales transactions from both channels.

5. SQL Business Analysis

SQL was used to perform:

- Overall sales KPI analysis

- Total orders and sales-line analysis
- Gross profit and gross profit margin analysis
- Sales channel comparison
- Territory-wise sales analysis
- Average order value analysis
- Customer and product validation
- Product category profitability analysis
- Date-based sales analysis
- Foreign-key integrity validation

6. Power BI

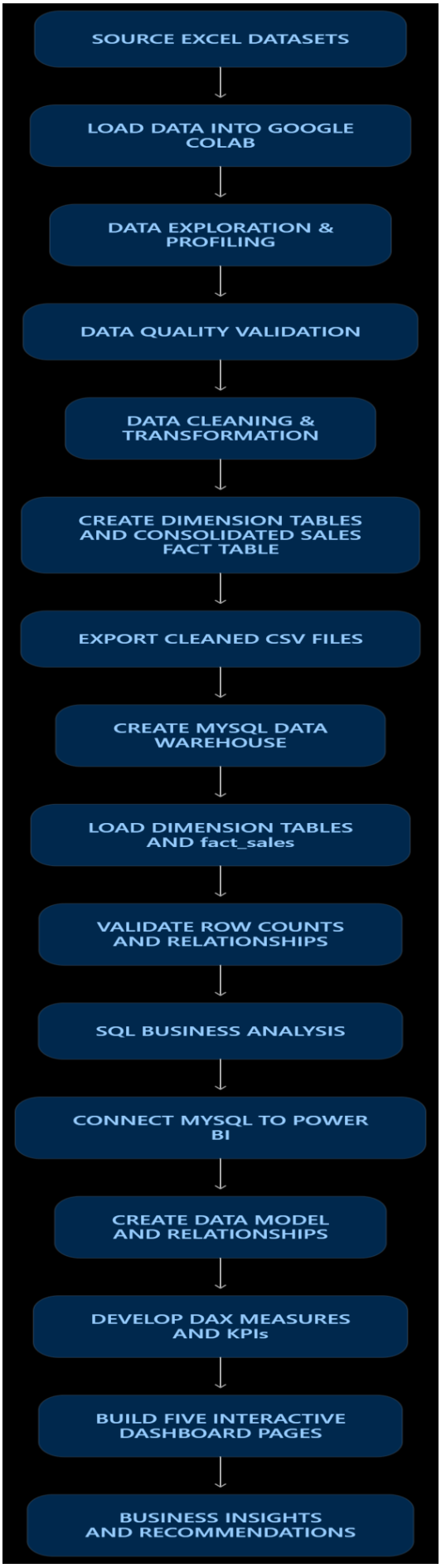
Power BI Desktop was used for:

- Connecting to the MySQL Data Warehouse
- Establishing relationships between fact and dimension tables
- Creating DAX measures
- Developing KPI cards
- Building bar, column, donut, line, and ribbon charts
- Creating interactive slicers
- Comparing sales channels, territories, customer segments, age groups, and product categories
- Presenting business insights through five dashboard pages

The Power BI solution contains:

1. Sales Overview
2. Territory Analysis
3. Customer Segmentation
4. Age Group Analysis
5. Product Profitability

2.2 Architecture Flow



III. PROJECT APPROACH

3.1 Methodology

Three source workbooks were identified:

- Dimension Tables
- Internet Sales
- Reseller Sales

The source files were reviewed to understand their structure, columns, identifiers, dates, and business meaning.

Initial Dataset Size

The source dimension tables included customer, employee, geography, product, reseller, and sales territory data. The two sales datasets contained Internet and Reseller transactions.

Dataset	Final records	Columns
dim_customer	18,484	15
dim_employee	21	9
dim_geography	655	6
dim_product	397	7
dim_reseller	701	4
dim_sales_territory	11	4
dim_date	1,127	9
fact_sales	121,253	15

Data Extraction & Exploration

The datasets were loaded into Google Colab using Python and Pandas.

The analysis included:

- Record counts
- Column names and data types
- Missing-value analysis
- Duplicate analysis
- Unique-key analysis
- Numerical statistics
- Date-range analysis
- Channel-wise record counts
- Referential integrity checks

Data Validation

Validation checks were performed for:

- Duplicate sales records
- Unique dimension identifiers
- Missing and invalid keys
- Date fields
- Product, customer, reseller, employee, and territory references
- Channel-specific null values
- Consolidated sales row counts

The final fact_sales table contained 121,253 sales lines, comprising 60,398 Internet sales lines and 60,855 Reseller sales lines.

Three blank Reseller records and 41 Reseller records with invalid foreign keys were removed during preprocessing.

The final MySQL foreign-key validation returned zero invalid references for CustomerKey, EmployeeKey, OrderDate, ProductKey, ResellerKey, and SalesTerritoryKey.

Data Transformation

The transformation process included:

- Preparing consistent dimension tables
- Creating the date dimension
- Cleaning and validating sales records
- Consolidating Internet and Reseller sales
- Creating the SalesChannel field to distinguish the two channels
- Retaining channel-specific nulls where the source business process does not use a particular key
- Removing CarrierTrackingNumber from the final fact table because it was entirely blank and did not add analytical value

The final fact table contains sales amounts, product costs, discounts, freight, order identifiers, order line numbers, transaction dates, dimensional keys, and sales channel information.

Data Staging & Loading

The transformed datasets were exported as CSV files and loaded into MySQL Workbench.

The final files included:

- dim_customer.csv
- dim_employee.csv
- dim_geography.csv
- dim_product.csv
- dim_reseller.csv
- dim_sales_territory.csv
- dim_date.csv
- fact_sales.csv

A staging table named fact_sales_staging was used during loading. It was excluded from the final Power BI model.

MySQL Data Warehouse

The cleaned analytical tables were loaded into:

adventure_works_dw

The warehouse uses a fact-and-dimension structure, with fact_sales connected to the appropriate dimensions through keys.

Power BI Data Modeling & Reporting

The MySQL Data Warehouse was connected to Power BI Desktop. The model includes active relationships between the sales fact table and customer, product, reseller, employee, sales territory, and date dimensions. Geography is linked to customer and reseller dimensions. The geography-to-reseller relationship was made inactive to avoid ambiguous filter paths in the model.

The dashboards were developed using DAX measures, KPI cards, charts, and slicers.

3.2 Technology Stack

Component	Technology	Purpose
Source data	Excel	Adventure Works sales and dimension data
Data processing	Python, Pandas	Exploration, cleaning, transformation
Development environment	Google Colab	Python-based ETL and analysis
Data staging	CSV	Transfer of transformed tables
Data Warehouse	MySQL	Structured analytical storage
Database tool	MySQL Workbench	Database creation, data loading, SQL analysis
Business analysis	SQL	Sales and profitability analysis
Visualization	Power BI Desktop	Interactive dashboard development
Calculations	DAX	KPI and analytical measures
Documentation	Microsoft Word	Case study reporting

IV. SOURCE DATA & TABLE MAPPING

4.1 Source Data

The project uses three primary source workbooks.

Source file	Purpose
`DimTables.xlsx`	Customer, employee, geography, product, reseller, and territory dimensions
`FactInternetSales.xlsx`	Direct Internet sales transactions
`FactResellerSales.xlsx`	Business-to-business Reseller sales transactions

```
DimTables sheets:
['DimCustomer', 'DimEmployee', 'DimGeography', 'DimProduct', 'DimReseller', 'DimSalesTerritory']

FactInternetSales sheets:
['FactInternetSales']

FactResellerSales sheets:
['FactResellerSales']
```

4.2 Data loaded In Google Colab

```
All tables loaded successfully.

Dimension Tables:
DimCustomer: (18484, 13)
DimEmployee: (21, 9)
DimGeography: (655, 6)
DimProduct: (397, 7)
DimReseller: (701, 4)
DimSalesTerritory: (11, 4)

Fact Tables:
FactInternetSales: (60398, 13)
FactResellerSales: (60899, 14)
```

V. DATA EXPLORATION & PRIMARY KEY VALIDATION

5.1 Data Exploration

Initial data profiling was performed using Python and Pandas.

The analysis included:

- Record counts and column names
- Data-type inspection
- Missing-value analysis
- Duplicate analysis
- Unique identifier validation
- Numerical statistics
- Date validation
- Channel-level analysis
- Dimension and fact-table consistency checks

The final dataset sizes were verified against the records loaded into MySQL.

Dataset	Records
Sales Data	5,000
Purchase Orders	1000
Inventory Data	200
Supplier Data	50

5.2 Dimension and Fact Record Validation

Table	Records
dim_customer	18,484
dim_employee	21
dim_geography	655
dim_product	397
dim_reseller	701
dim_sales_territory	11
dim_date	1,127
fact_sales	121,253

5.3 Sales Analysis

The consolidated sales fact table contained:

- 121,253 sales lines
- 31,455 distinct sales orders
- 60,398 Internet sales lines
- 60,855 Reseller sales lines
- No duplicate order lines in the final cleaned dataset
- Zero invalid references in the verified foreign-key checks

Channel-specific null values were retained where appropriate:

- Internet sales have customer keys and do not require reseller or employee keys in this model.
- Reseller sales use reseller and employee keys and do not use customer keys.

The absence of a quantity field means the number of sales lines should not be interpreted as units sold.

VI. MYSQL DATA WAREHOUSE IMPLEMENTATION & DATA LOADING

6.1 MySQL DataBase Implementation

After data cleaning and transformation, the analytical tables were loaded into the MySQL database:

adventure_works_dw

MySQL Workbench was used for database creation, table loading, SQL analysis, record-count validation, and referential integrity checks.



Figure 6.1: Tables created in MySQL Workbench

6.2 Data Warehouse Structure

- **Dim_customer**

CustomerKey	GeographyKey	CustomerName	BirthDate	MaritalStatus	Gender	EmailAddress	YearlyIncome	Education	Occupation	HouseOwnerFlag	Address	FirstPurchaseDate	Age	AgeGroup
11000	26	Jon Yang	1971-10-06	M	M	jon24@adventure-works.com	90000.00	Bachelors	Professional	1	3761 N. 14th St	2011-01-19	42	40-45
11001	37	Eugene Huang	1976-05-10	S	M	eugene10@adventure-works.com	60000.00	Bachelors	Professional	0	2243 W St.	2011-01-15	37	35-40
11002	31	Ruben Torres	1971-02-09	M	M	ruben35@adventure-works.com	60000.00	Bachelors	Professional	1	5844 Linden Land	2011-01-07	42	40-45
11003	11	Christy Zhu	1973-08-14	S	F	christy12@adventure-works.com	70000.00	Bachelors	Professional	0	1825 Village Pl.	2010-12-29	40	40-45

- **Dim_date**

Date	Year	Quarter	MonthNumber	MonthName	YearMonth	Day	DayName	WeekNumber
2010-12-29	2010	Q4	12	December	2010-12	29	Wednesday	52
2010-12-30	2010	Q4	12	December	2010-12	30	Thursday	52
2010-12-31	2010	Q4	12	December	2010-12	31	Friday	52
2011-01-01	2011	Q1	1	January	2011-01	1	Saturday	52

- **Dim_employee**

EmployeeKey	ParentEmployeeKey	SalesTerritoryKey	EmployeeName	Title	EmailAddress	DepartmentName	HireDate	BirthDate
112	NULL	11	Ken Sánchez	Chief Executive Officer	ken0@adventure-works.com	Executive	2008-08-14	1968-08-29
143	112	11	Laura Norman	Chief Financial Officer	laura1@adventure-works.com	Executive	2008-08-31	1975-08-06
144	112	11	Laura Norman	Chief Financial Officer	laura1@adventure-works.com	Executive	2008-08-31	1975-08-06
272	277	11	Stephen Jiang	North American Sales Manager	stephen0@adventure-works.com	Sales	2010-08-04	1951-05-17
277	112	11	Brian Welcker	Vice President of Sales	brian3@adventure-works.com	Sales	2010-09-15	1977-01-04

- **Dim_geography**

GeographyKey	City	State	Country	PostalCode	SalesTerritoryKey
1	Alexandria	New South Wales	Australia	2015	9
2	Coffs Harbour	New South Wales	Australia	2450	9
3	Darlinghurst	New South Wales	Australia	2010	9
4	Goulburn	New South Wales	Australia	2580	9
5	Lane Cove	New South Wales	Australia	1597	9

- **Dim_product**

ProductKey	ProductSubcategoryKey	Product	Color	Model	Subcategory	Category
210	14	HL Road Frame - Black, 58	Black	HL Road Frame	Road Frames	Components
211	14	HL Road Frame - Red, 58	Red	HL Road Frame	Road Frames	Components
212	31	Sport-100 Helmet, Red	Red	Sport-100	Helmets	Accessories
213	31	Sport-100 Helmet, Red	Red	Sport-100	Helmets	Accessories
214	31	Sport-100 Helmet, Red	Red	Sport-100	Helmets	Accessories

- **Dim_reseller**

SalesKey	ProductKey	CustomerKey	ResellerKey	EmployeeKey	SalesTerritoryKey	SalesOrderNumber	SalesOrderLineNumber	DiscountAmount	TotalProductCost	SalesAmount	Freight	CarrierTrackingNumber	OrderDate	DueDate	ShipDate	SalesChannel
1	310	21768	NULL	NULL	6	SO43697	1	0.0000	2171.2942	3578.2700	89.4568	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
2	346	28389	NULL	NULL	7	SO43698	1	0.0000	1912.1544	3399.9900	84.9998	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
3	346	25863	NULL	NULL	1	SO43699	1	0.0000	1912.1544	3399.9900	84.9998	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
4	336	14501	NULL	NULL	4	SO43700	1	0.0000	413.1463	699.0982	17.4775	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
5	346	11003	NULL	NULL	9	SO43701	1	0.0000	1912.1544	3399.9900	84.9998	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
6	311	27645	NULL	NULL	4	SO43702	1	0.0000	2171.2942	3578.2700	89.4568	NULL	2010-12-30	2011-01-11	2011-01-06	Internet
7	310	16624	NULL	NULL	9	SO43703	1	0.0000	2171.2942	3578.2700	89.4568	NULL	2010-12-30	2011-01-11	2011-01-06	Internet

- **Dim_sales_territory**

SalesTerritoryKey	SalesTerritoryRegion	SalesTerritoryCountry	SalesTerritoryGroup
1	Northwest	United States	North America
2	Northeast	United States	North America
3	Central	United States	North America
4	Southwest	United States	North America
5	Southeast	United States	North America

- **Fact_sales**

SalesKey	ProductKey	CustomerKey	ResellerKey	EmployeeKey	SalesTerritoryKey	SalesOrderNumber	SalesOrderLineNumber	DiscountAmount	TotalProductCost	SalesAmount	Freight	CarrierTrackingNumber	OrderDate	DueDate	ShipDate	SalesChannel
1	310	21768	NULL	NULL	6	SO43697	1	0.0000	2171.2942	3578.2700	89.4568	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
2	346	28389	NULL	NULL	7	SO43698	1	0.0000	1912.1544	3399.9900	84.9998	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
3	346	25863	NULL	NULL	1	SO43699	1	0.0000	1912.1544	3399.9900	84.9998	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
4	336	14501	NULL	NULL	4	SO43700	1	0.0000	413.1463	699.0982	17.4775	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
5	346	11003	NULL	NULL	9	SO43701	1	0.0000	1912.1544	3399.9900	84.9998	NULL	2010-12-29	2011-01-10	2011-01-05	Internet
6	311	27645	NULL	NULL	4	SO43702	1	0.0000	2171.2942	3578.2700	89.4568	NULL	2010-12-30	2011-01-11	2011-01-06	Internet
7	310	16624	NULL	NULL	9	SO43703	1	0.0000	2171.2942	3578.2700	89.4568	NULL	2010-12-30	2011-01-11	2011-01-06	Internet

VII. POWERBI DATA MODEL & KPI DEVELOPMENT

7.1 Power BI Data Model

The data model was developed in Power BI Desktop using the cleaned dimension tables and the consolidated fact_sales table imported from the MySQL data warehouse. The model follows a dimensional data modelling approach, with the sales fact table connected to the relevant dimension tables to support interactive analysis of sales performance, customer behaviour, product profitability, and sales territories.



Figure 7.1: Power BI Data Model

7.2 POWER BI RELATIONSHIPS

The relationships were configured with appropriate cardinality and filter direction to support accurate reporting.

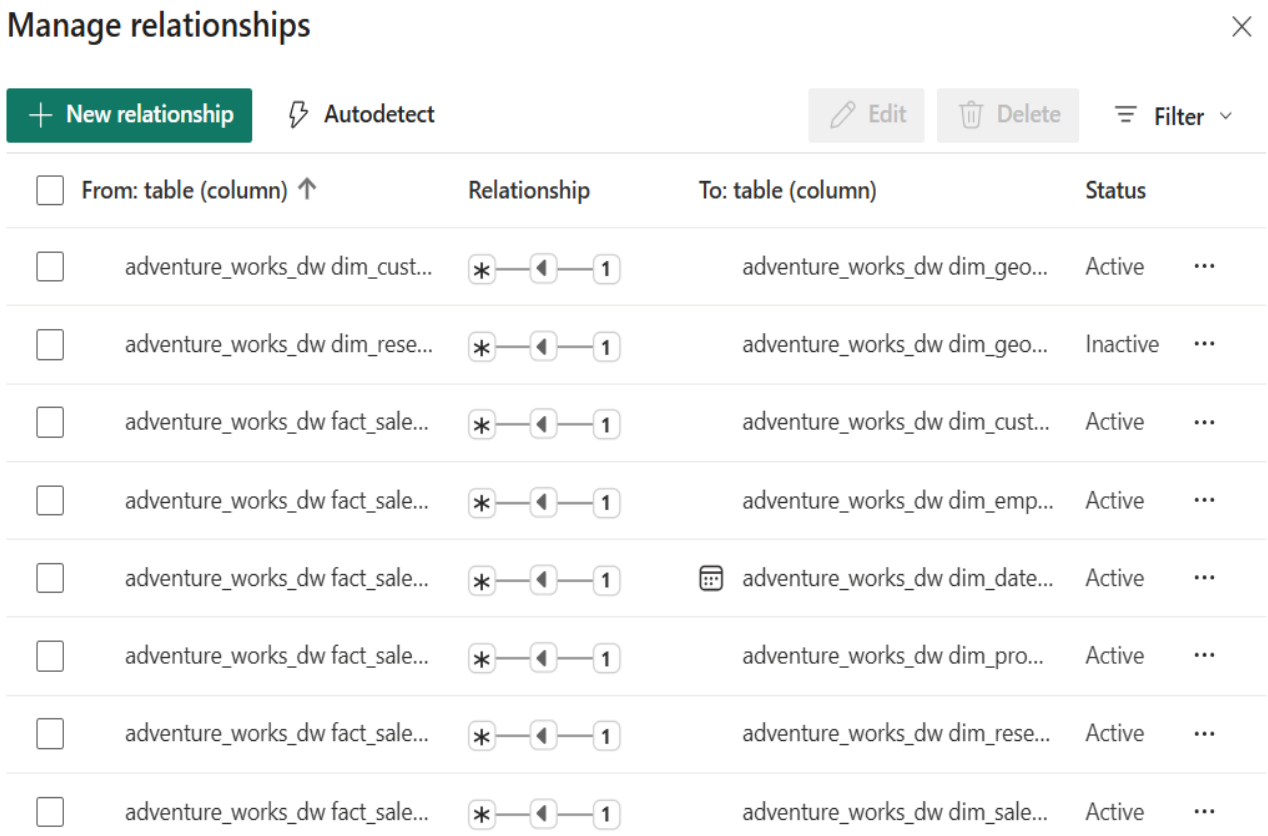


Figure 7.2: Power BI Table Relationships

VIII. Power BI Dashboards & Key Insights

8.1 Sales Overview Dashboard

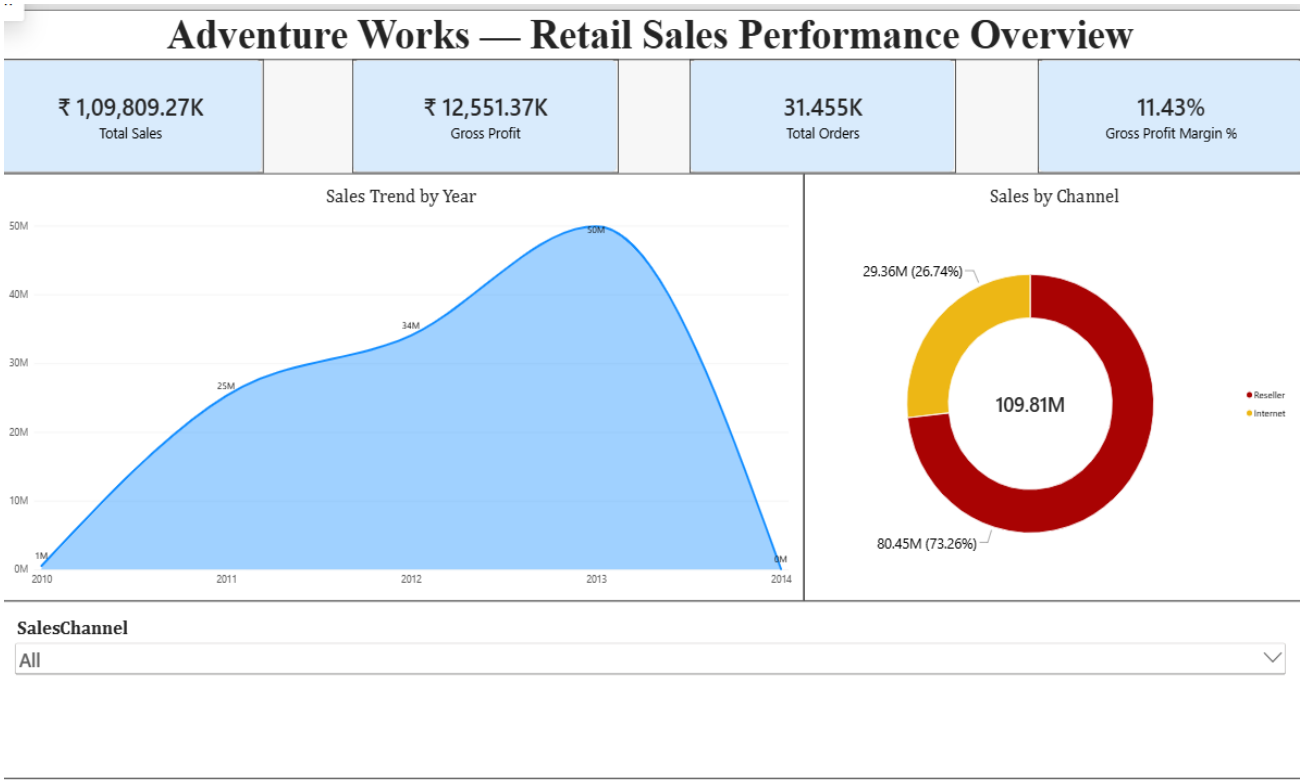


Figure 8.1: Sales Overview Dashboard

Key Insights:

- Total sales reached approximately **$109.81 million**, with gross profit of approximately **$12.55 million**.
- The reseller channel contributed the majority of revenue, while the Internet channel generated a substantially higher gross profit margin.
- Annual sales increased from approximately $25.27 million in 2011 to $49.93 million in 2013, indicating strong growth during this period.

8.2 Territory Analysis Dashboard

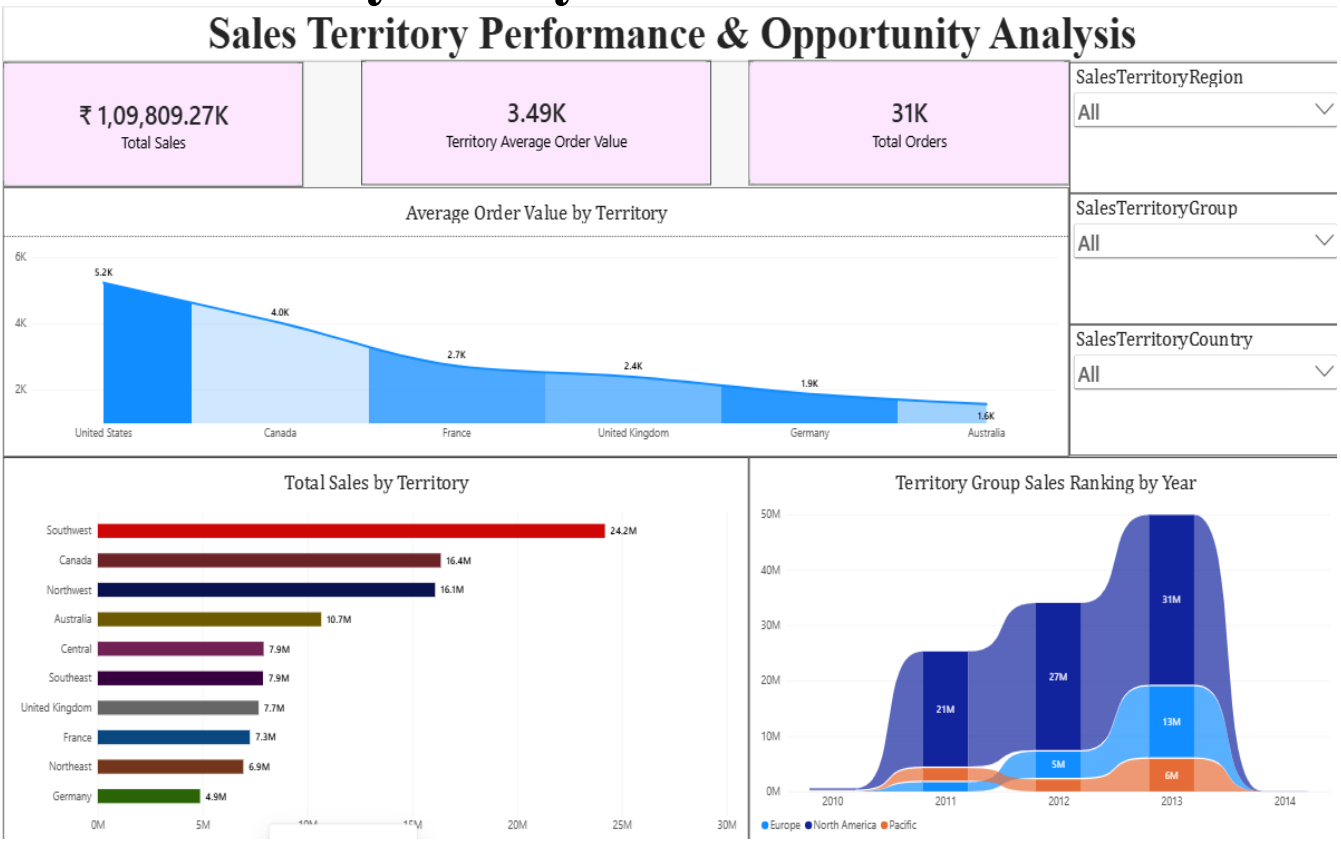


Figure 8.2: Sales Territory Performance & Opportunity Analysis Dashboard

Key Insights:

- The Southwest territory recorded the highest sales, at approximately $24.18 million.
- Canada and Northwest were also major contributors to total sales.
- Southeast, Northeast, and Central recorded comparatively high average order values, indicating differences in order value across territories.

8.3 Customer Segmentation Dashboard

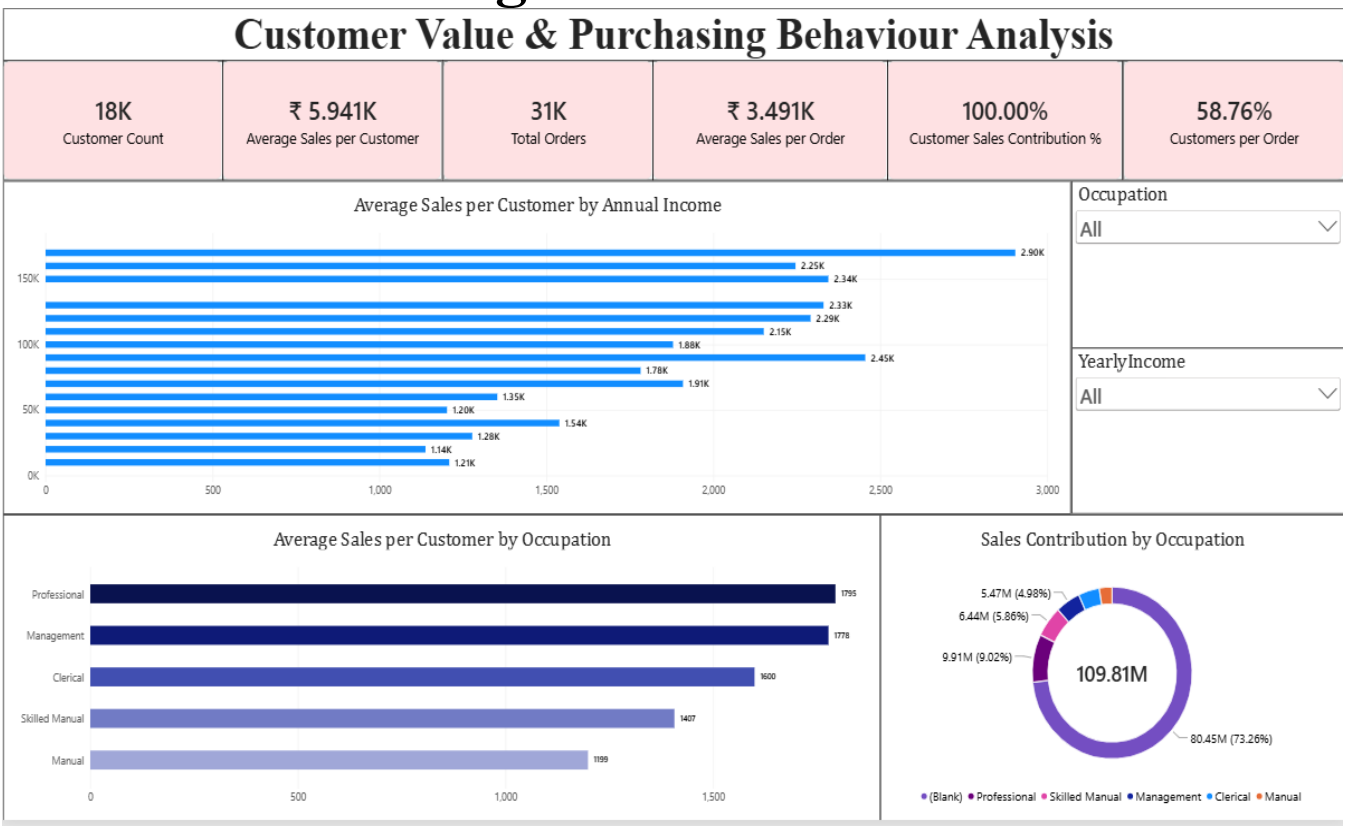


Figure 8.3: Customer Value & Purchasing Behaviour Analysis Dashboard

Key Insights:

- Average sales per customer varied across occupational groups, with Professional customers recording approximately $1,794.92.
- Customers in higher annual income groups generally showed higher average sales per customer.
- Segmenting customers by occupation and income helps identify customer groups for targeted marketing and sales strategies.

8.4 Age Group Analysis Dashboard

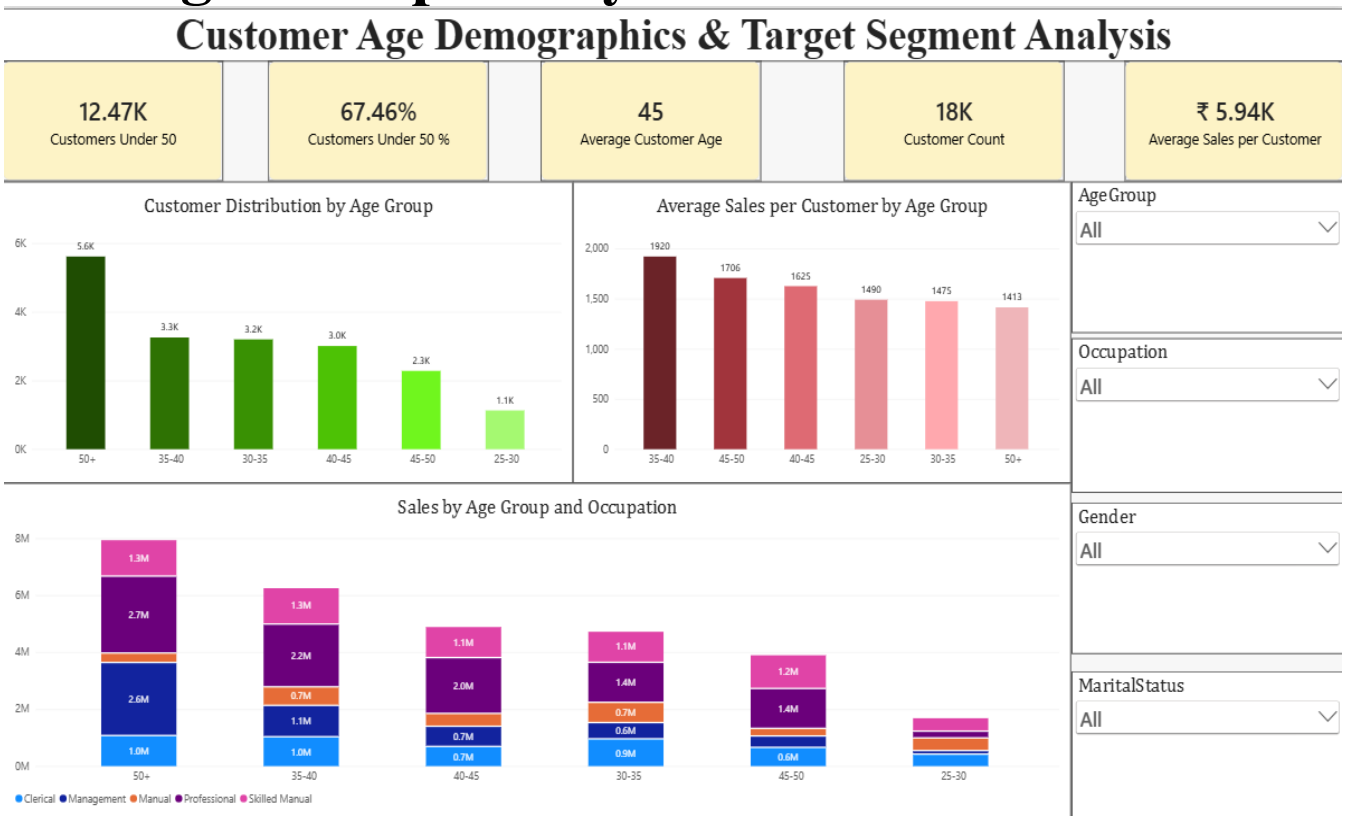


Figure 8.4: Customer Age Demographics & Target Segment Analysis Dashboard

Key Insights:

- Customers below 50 years of age represented approximately **69.63%** of the customer base.
- The 50+ age group was the largest age segment, with 5,613 customers.
- The 35–40 age group recorded the highest average sales per customer among the analysed age groups, at approximately $1,919.71.

8.5 Product Profitability Dashboard

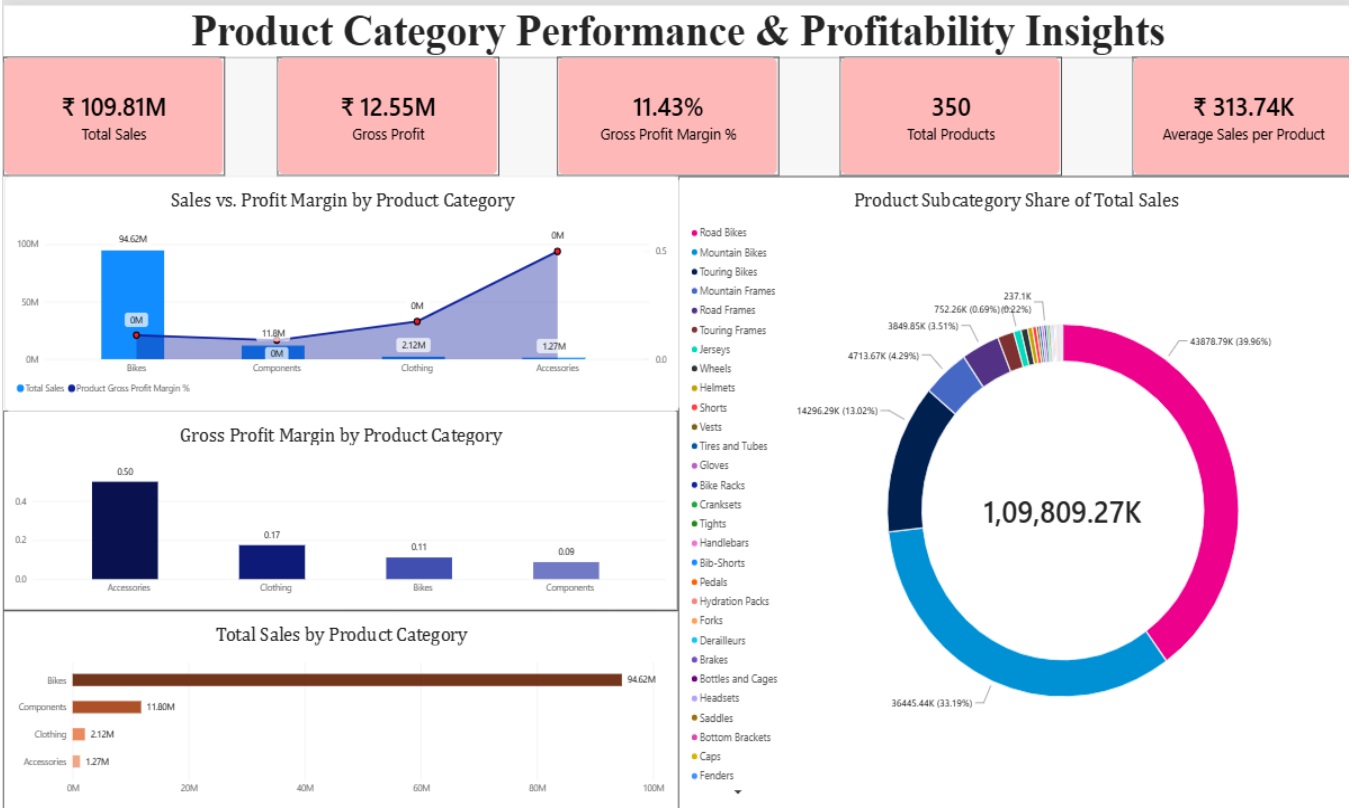


Figure 8.5: Product Category Performance & Profitability Insights Dashboard

Key Insights:

- The Bikes category contributed approximately **86.17% of total sales**, making it the dominant product category.
- Accessories recorded the highest gross profit margin, at approximately 49.88%, despite contributing a small share of total revenue.
- Clothing also recorded a higher gross profit margin than Bikes and Components, highlighting differences in profitability across product categories.

IX. CONCLUSION

9.1 Conclusion

The Data-Driven Retail Sales Optimization case study developed an end-to-end sales analytics solution using Python, Google Colab, MySQL Workbench, SQL, Power BI, and DAX.

The project consolidated Internet and Reseller sales into a structured MySQL Data Warehouse supported by customer, employee, geography, product, reseller, sales territory, and date dimensions. Data validation and relational checks helped establish a consistent foundation for analysis.

The Power BI dashboards provide an interactive view of sales trends, channel performance, territory comparisons, customer segmentation, age-group patterns, and product profitability. The resulting analysis highlights differences in revenue contribution, order value, customer spending patterns, and gross profit margins.

The solution demonstrates how structured ETL, dimensional modeling, SQL analysis, and business intelligence reporting can transform raw sales datasets into actionable business insights.

9.2 Key Outcomes

- Implemented a sales analytics workflow using Google Colab, Python, MySQL, SQL, and Power BI.
- Created a structured MySQL Data Warehouse containing seven dimension tables and one consolidated sales fact table.
- Consolidated 60,398 Internet sales lines and 60,855 cleaned Reseller sales lines into 121,253 sales lines.
- Validated dimension records, foreign-key relationships, and overall sales totals.
- Developed DAX measures for sales, gross profit, gross profit margin, orders, channel contribution, customer segmentation, age analysis, and product profitability.
- Built five Power BI dashboard pages for sales overview, territory analysis, customer segmentation, age-group analysis, and product profitability.
- Identified the Reseller channel's dominant revenue contribution and comparatively low gross profit margin.
- Identified Southwest as the highest-revenue territory.
- Identified customer occupation and income groups with higher average sales.

- Analyzed the under-50 customer segment, which represents approximately 69.63% of customers.
- Identified Bikes as the dominant revenue category and Accessories as the category with the highest gross profit margin.
- Produced business recommendations for territory development, customer targeting, channel profitability, and product optimization.

9.3 GitHub Link

https://github.com/surabhi-0405/Case-Study-7-Data-Driven-Retail-Sales-Optimization-using-Adventure-Works